

18B.15. Diffusion and heterogeneous reaction in a slender cylindrical tube with a closed end (Fig. 18B.15). A slender cylindrical pore of length L , cross-sectional area S , and perimeter P , is in contact at its open end with a large body of well-mixed fluid, consisting of species A and B . Species A , a minor constituent of this fluid, disappears into the pore, diffuses in the z direction and reacts on its walls. The rate of this reaction may be expressed as $(n \cdot n_s)_{\text{surface}} = f(\omega_A)$; that is, at the wall the mass flux normal to the surface is some function of the mass fraction, ω_A , of A in the fluid adjacent to the solid surface. The mass fraction ω_A depends on z , the distance from the inlet. Because A is present in low concentration, the fluid temperature and density may be considered constant, and the diffusion flux is adequately described by $J_A = -D_{AB} \nabla \omega_A$.

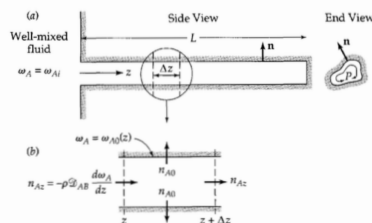


Fig. 18B.15. (a) Diffusion and heterogeneous reaction in a long, non-circular cylinder. (b) Region of thickness Δz over which the mass balance is made.

where the diffusivity may be regarded as a constant. Because the pore is long compared to its lateral dimension, concentration gradients in the lateral directions may be neglected. Note the similarity with the problem discussed in §10.7.

(a) Show by means of a shell balance that, at steady state,

$$-\frac{dN_{A,z}}{dz} = \frac{P}{S} f(\omega_A) \quad (18B.15-1)$$

(b) Show that the steady-state mass average velocity v_z is zero for this system.

(c) Substitute the appropriate form of Fick's law into Eq. 18B.15-1, and integrate the resulting differential equation for the special case that $f(\omega_A) = k_1' \omega_A$. To obtain a boundary condition at $z = L$, neglect the rate of reaction on the closed end of the cylinder; why is this a reasonable approximation?

(d) Develop an expression for the total rate $\dot{W}_A$ of disappearance of A in the cylinder.

(e) Compare the results of parts (c) and (d) with those of §10.7 both from the standpoint of the mathematical development and the nature of the assumptions made.

Answers: (c) $\frac{\omega_A}{\omega_{A,i}} = \frac{\cosh N[1 - (z/L)]}{\cosh N}$, where $N = \sqrt{\frac{PL^2 k_1'}{SgD_{AB}}}$; (d) $\dot{W}_A = (SgD_{AB} \omega_{A,i} / L) N \tanh N$

$$a.) \quad \left([S N_{A,z}]_z - [S N_{A,z}]_{z+\Delta z} - f(\omega_A) P \Delta z = 0 \right) \frac{1}{S \Delta z}$$

$$\begin{aligned} & [N_{A,z}]_z - [N_{A,z}]_{z+\Delta z} - \frac{P}{S} f(\omega_A) \Delta z = 0 \\ & - \frac{dN_{A,z}}{dz} = \frac{P}{S} f(\omega_A) \end{aligned}$$

$$b.) \quad \omega_A \ll 1$$

$$N_{B,z} = 0 \quad (\text{No B diffuses})$$

$$N_z = g v_z = N_{A,z} + N_{B,z}$$

$$g v_z = N_{A,z}$$

$$N_{A,z} = g \omega_A v_z + J_{A,z}$$

$$g v_z = g \omega_A v_z + J_{A,z}$$

$$v_z - \omega_A v_z = \frac{1}{g} J_{A,z}$$

$$v_z (1 - \omega_A) = \frac{1}{g} J_{A,z}$$

$$v_z = \frac{J_{A,z}}{g(1 - \omega_A)} = \frac{-g D_{AB} \nabla \omega_A}{g(1 - \omega_A)}$$

$$v_z = \lim_{\omega_A \rightarrow 0} \frac{-g D_{AB} \nabla \omega_A}{g(1 - \omega_A)} = 0$$

(c.)

$$N_{A,z} = g \omega_A v_z - g D_{AB} \frac{d\omega_A}{dz}$$

$$N_{A,z} = -g D_{AB} \frac{d\omega_A}{dz}$$

$$-\frac{dN_{A,z}}{dz} = \frac{P}{S} f(\omega_A)$$

$$-(-g D_{AB}) \frac{d}{dz} \left(\frac{d\omega_A}{dz} \right) = \frac{P}{S} f(\omega_A)$$

$$g D_{AB} \frac{d^2 \omega_A}{dz^2} = \frac{P}{S} f(\omega_A)$$

$$g D_{AB} \frac{d^2 \omega_A}{dz^2} = \frac{P}{S} k_1' \omega_A$$

$$\frac{d^2 \omega_A}{dz^2} = \frac{P k_1'}{S g D_{AB}} \omega_A$$

$$\frac{d^2 \omega_A}{dz^2} = \mu^2 \omega_A$$

$$\frac{d^2 \omega_A}{dz^2} - \mu^2 \omega_A = 0$$

$$f(\omega_A) = k_1' \omega_A \quad (\text{from problem})$$

$$\omega_A(z) = C_1 \cosh(\mu z) + C_2 \sinh(\mu z)$$

$$\frac{d\omega_A}{dz} = \mu C_1 \sinh(\mu z) + \mu C_2 \cosh(\mu z)$$

$$@ \omega_A(0) = \omega_{A,i}$$

$$\omega_{A,i} = C_1 (\cosh(\mu(0)) + C_2 \sinh(\mu(0)))$$

$$\omega_{A,i} = C_1 (\cosh(0) + C_2 \sinh(0))$$

$$\omega_{A,i} = C_1 (1) + C_2 (0)$$

$$\omega_{A,i} = C_1$$

$$@ \frac{d\omega_A}{dz} \bigg|_{z=L} = 0$$

$$0 = \mu (C_1 \sinh(\mu L) + \mu C_2 \cosh(\mu L))$$

$$- \mu C_1 \sinh(\mu L) = \mu C_2 \cosh(\mu L)$$

$$C_2 = -C_1 \frac{\sinh(\mu L)}{\cosh(\mu L)} = -C_1 \tanh(\mu L)$$

$$C_2 = -\omega_{A,i} \tanh(\mu L)$$

$$\omega_A(z) = \omega_{A,i} \cosh(\mu z) - \omega_{A,i} \tanh(\mu L) \sinh(\mu z)$$

$$\omega_A(z) = \omega_{A,i} \cosh(\mu z) - \omega_{A,i} \frac{\sinh(\mu L) \sinh(\mu z)}{\cosh(\mu L)}$$

$$\omega_A(z) = \omega_{A,i} \left[\frac{\cosh(\mu z) \cosh(\mu L) - \sinh(\mu L) \sinh(\mu z)}{\cosh(\mu L)} \right]$$

$$\cosh(x-y) = \cosh(x) \cosh(y) - \sinh(x) \sinh(y)$$

$$\omega_A(z) = \omega_{A,i} \left[\frac{\cosh(\mu(L-z))}{\cosh(\mu L)} \right] = \omega_{A,i} \frac{\cosh(\mu(L-z))}{\cosh(\mu L)}$$

(Thiele Modulus)

$$N = \mu L \Rightarrow N = L \sqrt{\frac{P k_1'}{S g D_{AB}}} = \sqrt{\frac{P k_1' L^2}{S g D_{AB}}}$$

$$\mu = \frac{N}{L}$$

$$\frac{\omega_A(z)}{\omega_{A,i}} = \frac{\cosh[N(1 - \frac{z}{L})]}{\cosh(N)} \quad \text{where } N = \sqrt{\frac{P k_1' L^2}{S g D_{AB}}}$$

d.)

$$\text{mass flux} \rightarrow W_A = S N_A z = -S_g D_{AB} \left. \frac{dw_A}{dz} \right|_{z=0}$$

↳ into cylinder

$$\frac{w_A(z)}{w_{A1}} = \frac{\cosh\left[N\left(1-\frac{z}{L}\right)\right]}{\cosh(N)}$$

$$\frac{d}{dz} \left(w_A(z) = w_{A1} \frac{\cosh\left[N\left(1-\frac{z}{L}\right)\right]}{\cosh(N)} \right)$$

$$\frac{dw_A}{dz} = w_{A1} \frac{\sinh\left[N\left(1-\frac{z}{L}\right)\right] \left(-\frac{1}{L}\right)}{\cosh(N)}$$

$$\left. \frac{dw_A}{dz} \right|_{z=0} = \frac{w_{A1} \sinh\left[N\left(1-\frac{0}{L}\right)\right] \left(-\frac{1}{L}\right)}{\cosh(N)}$$

$$\left. \frac{dw_A}{dz} \right|_{z=0} = \frac{-w_{A1} N \sinh(N)}{L \cosh(N)} = \frac{-w_{A1} N}{L} \tanh(N)$$

$$W_A = -S_g D_{AB} \left. \frac{dw_A}{dz} \right|_{z=0}$$

$$W_A = -S_g D_{AB} \left(\frac{-w_{A1} N}{L} \right) \tanh(N)$$

$$W_A = \left(\frac{S_g D_{AB} w_{A1}}{L} \right) N \tanh(N)$$

e.) The problem is similar to the cooling fin problem 10.7

18C.3. Absorption rate in a falling film. The result in Eq. 18.5-18 may be obtained by an alternative procedure.

(a) According to an overall mass balance on the film, the total moles of A transferred per unit time across the gas-liquid interface must be the same as the total molar rate of flow of A across the plane $z = L$. The latter rate is calculated as follows:

$$W_A = \lim_{\delta \rightarrow \infty} (W \delta v_{\max}) \left(\frac{1}{\delta} \int_0^\delta c_A|_{z=L} dx \right) = W v_{\max} \int_0^\infty c_A|_{z=L} d\xi \quad (18C.3-1)$$

Explain this procedure carefully.

(b) Insert the solution for c_A in Eq. 18.5-15 into the result of (a) to obtain:

$$W_A = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \int_0^\infty \left(\int_0^\infty \frac{x}{\sqrt{4D_{AB}z}} \exp(-\xi^2) d\xi \right) d\xi$$

$$= W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \sqrt{\frac{4D_{AB}L}{v_{\max}}} \int_0^\infty \left(\int_0^\infty \exp(-\xi^2) d\xi \right) d\xi \quad (18C.3-2)$$

In the second line, the new variable $u = x/\sqrt{4D_{AB}L/v_{\max}}$ has been introduced.

(c) Change the order of integration in the double integral, to get

$$W_A = W L c_{A0} \sqrt{\frac{4D_{AB}v_{\max}}{\pi L}} \cdot 2 \int_0^\infty \exp(-\xi^2) \left(\int_0^\xi du \right) d\xi \quad (18C.3-3)$$

a.)

There is conservation of mass for the film of length L . The rate of A through the gas-liquid interface has to be the same as the A that leaves the system at the bottom (when $z=L$).

For a thin film, the velocity is approximated to its maximum value due to the short interaction time between the film and interface.

The total mol flow rate when $z=L$ is found by integrating the mole flux across the film of thickness δ . Species does not go far into the film so the limit is taken as $\delta \rightarrow \infty$.

$$\text{Thus all leads to the equation: } W_A = \lim_{\delta \rightarrow \infty} (W \delta v_{\max}) \left(\frac{1}{\delta} \int_0^\delta c_A|_{z=L} dx \right) = W v_{\max} \int_0^\infty c_A|_{z=L} d\xi$$

b.)

$$\frac{c_A}{c_{A0}} = 1 - \frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{4D_{AB}z/v_{\max}}} \exp(-\xi^2) d\xi \quad (18.5-15)$$

$$\frac{c_A}{c_{A0}} = 1 - \operatorname{erfc} \left(\frac{x}{\sqrt{4D_{AB}z/v_{\max}}} \right) = \operatorname{erfc} \left(\frac{x}{\sqrt{4D_{AB}L/v_{\max}}} \right) = \frac{2}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4D_{AB}L/v_{\max}}}}^\infty \exp(-s^2) ds$$

At $z=L$:

$$\frac{c_A}{c_{A0}} = \frac{2}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4D_{AB}L/v_{\max}}}}^\infty \exp(-s^2) ds \Rightarrow c_A = c_{A0} \frac{2}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4D_{AB}L/v_{\max}}}}^\infty \exp(-s^2) ds$$

$$W_A = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \int_0^\infty \left(\int_{\frac{x}{\sqrt{4D_{AB}L/v_{\max}}}}^\infty \exp(-s^2) ds \right) dx$$

$$u = \frac{x}{\sqrt{4D_{AB}L/v_{\max}}}$$

$$du = \frac{dx}{\sqrt{4D_{AB}L/v_{\max}}} \Rightarrow dx = du \sqrt{4D_{AB}L/v_{\max}}$$

$$W_A = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \left(\sqrt{\frac{4D_{AB}L}{v_{\max}}} \right) \int_0^\infty \left(\int_0^\infty \exp(-s^2) ds \right) du$$

c.) For a fixed ξ , u ranges from 0 to ξ
 ξ ranges from 0 to ∞

$$W_A = W c_{A0} \sqrt{\frac{4D_{AB}v_{\max}L}{\pi}} \cdot 2 \int_0^\infty \exp(-\xi^2) \left(\int_0^\xi du \right) d\xi$$

19B.1. **Steady-state evaporation** (Fig. 18.2-1). Rework the problem solved in §18.2, dealing with the evaporation of liquid A into gas B, starting from Eq. 19.1-17.

(a) First obtain an expression for v^* , using Eq. (M) of Table 17.8-1, as well as Fick's law in the form of Eq. (D) of Table 17.8-2.

(b) Show that Eq. 19.1-17 then becomes the following nonlinear second-order differential equation:

$$\frac{d^2 x_A}{dz^2} + \frac{1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 = 0 \quad (19B.1-1)$$

(c) Solve this equation to get the mole fraction profile given in Eq. 18.2-11.

Table 17.8-1 Notation for Mass and Molar Fluxes*

Quantity	With respect to stationary axes	With respect to mass average velocity v	With respect to molar average velocity v^*
Velocity of species a (cm/s)	v_a (A)	$v_a - v$ (B)	$v_a - v^*$ (C)
Mass flux of species a (g/cm ² s)	$n_a = \rho_a v_a$ (D)	$j_a = \rho_a (v_a - v)$ (E)	$j_a^* = \rho_a (v_a - v^*)$ (F)
Molar flux of species a (g-moles/cm ² s)	$N_a = c_a v_a$ (G)	$J_a = c_a (v_a - v)$ (H)	$J_a^* = c_a (v_a - v^*)$ (I)
Sums of mass fluxes	$\sum_{a=1}^N n_a = \rho v$ (J)	$\sum_{a=1}^N j_a = 0$ (K)	$\sum_{a=1}^N j_a^* = \rho(v - v^*)$ (L)
Sums of molar fluxes	$\sum_{a=1}^N N_a = cv^*$ (M)	$\sum_{a=1}^N J_a = c(v^* - v)$ (N)	$\sum_{a=1}^N J_a^* = 0$ (O)
Relations between mass and molar fluxes	$n_a = M_a N_a$ (P)	$j_a = M_a J_a$ (Q)	$j_a^* = M_a J_a^*$ (R)
Interrelations among mass fluxes	$n_a = j_a + \rho_a v$ (S)	$j_a = n_a - \omega_a \sum_{b=1}^N n_b$ (T)	$j_a^* = n_a - x_a \sum_{b=1}^N \frac{M_b}{M_a} n_b$ (U)
Interrelations among molar fluxes	$N_a = J_a^* + c_a v^*$ (V)	$J_a = N_a - \omega_a \sum_{b=1}^N \frac{M_b}{M_a} N_b$ (W)	$J_a^* = N_a - x_a \sum_{b=1}^N \frac{M_b}{M_a} N_b$ (X)

* Entries in the shaded boxes, involving the "hybrid fluxes" j_a^* and J_a^* , are seldom needed; they are included only for the sake of completeness.

Table 17.8-2 Equivalent Forms of Fick's (First) Law of Binary Diffusion

Flux	Gradient	Form of Fick's Law
j_A	$\nabla \omega_A$	$j_A = -\rho \mathcal{D}_{AB} \nabla \omega_A$ (A)
J_A^*	∇x_A	$J_A^* = -c \mathcal{D}_{AB} \nabla x_A$ (B)
n_A	$\nabla \omega_A$	$n_A = \omega_A (n_A + n_B) - \rho \mathcal{D}_{AB} \nabla \omega_A = \rho_A v - \rho \mathcal{D}_{AB} \nabla \omega_A$ (C)
N_A	∇x_A	$N_A = x_A (N_A + N_B) - c \mathcal{D}_{AB} \nabla x_A = c_A v^* - c \mathcal{D}_{AB} \nabla x_A$ (D)
$\rho(v_A - v_B)$	$\nabla \omega_A$	$\rho(v_A - v_B) = -\frac{\rho \mathcal{D}_{AB}}{\omega_A \omega_B} \nabla \omega_A$ (E)
$c(v_A - v_B)$	∇x_A	$c(v_A - v_B) = -\frac{c \mathcal{D}_{AB}}{x_A x_B} \nabla x_A$ (F)

$$c \left(\frac{\partial x_A}{\partial t} + (v^* \cdot \nabla) x_A \right) = c \mathcal{D}_{AB} \nabla^2 x_A + (x_B R_A - x_A R_B) \quad (19.1-17)$$

$$\left(\frac{1-x_A}{1-x_{A1}} \right) = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{z-z_1}{z_2-z_1}} \quad (18.2-11)$$

a.) $\sum N_a = N_A + N_B = c v^*$ (Only A diffuses)

$$N_A = c v^*$$

$$v^* = \frac{N_A}{c}$$

$$N_A = x_A (N_A + N_B) - c \mathcal{D}_{AB} \nabla x_A$$

$$N_A = x_A (N_A) - c \mathcal{D}_{AB} \frac{dx_A}{dz}$$

$$N_A - x_A N_A = -c \mathcal{D}_{AB} \frac{dx_A}{dz}$$

$$N_A = -\frac{c \mathcal{D}_{AB}}{(1-x_A)} \frac{dx_A}{dz}$$

$$v^* = \frac{1}{c} \left(-\frac{c \mathcal{D}_{AB}}{(1-x_A)} \frac{dx_A}{dz} \right)$$

$$v^* = -\frac{\mathcal{D}_{AB}}{(1-x_A)} \frac{dx_A}{dz}$$

$$b.) \quad c \left(\frac{\partial x_A}{\partial t} + (v^* \cdot \nabla) x_A \right) = c \mathcal{D}_{AB} \nabla^2 x_A + (x_B R_A - x_A R_B)$$

$$c v^* \frac{dx_A}{dz} = c \mathcal{D}_{AB} \frac{d^2 x_A}{dz^2}$$

$$c \left(\frac{-\mathcal{D}_{AB}}{(1-x_A)} \frac{dx_A}{dz} \right) \frac{dx_A}{dz} = c \mathcal{D}_{AB} \frac{d^2 x_A}{dz^2}$$

$$\frac{-1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 = \frac{d^2 x_A}{dz^2}$$

$$\frac{d^2 x_A}{dz^2} + \frac{1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 = 0$$

$$c.) \quad v = 1 - x_A$$

$$\frac{dv}{dz} = -\frac{dx_A}{dz}$$

$$\frac{d^2 v}{dz^2} = -\frac{d^2 x_A}{dz^2}$$

$$-\frac{d^2 v}{dz^2} + \frac{1}{v} \left(-\frac{dv}{dz} \right)^2 = 0$$

$$\frac{d^2 v}{dz^2} - \frac{1}{v} \left(\frac{dv}{dz} \right)^2 = 0$$

$$\frac{d^2 v}{dz^2} = \frac{1}{v} \left(\frac{dv}{dz} \right)^2$$

$$\left(\frac{d^2 v}{dz^2} \right) = \frac{1}{v} \left(\frac{dv}{dz} \right)^2$$

$$\left(\frac{dv}{dz} \right)$$

$$\int \frac{d}{dz} \left(\ln \left(\frac{dv}{dz} \right) \right) = \frac{d}{dz} \left(\ln(v) \right)$$

$$\ln \left(\frac{dv}{dz} \right) = \ln(v) + C_1$$

$$\ln \left(\frac{dv}{dz} \right) = \ln(v) + C_1$$

$$\frac{dv}{dz} = e^{\ln(v) + C_1} = e^{\ln(v)} \cdot e^{C_1} = v \cdot C_1$$

$$\frac{dv}{dz} = C_1 v$$

$$\frac{dv}{dz} = C_1 v$$

$$\int \frac{dv}{v} = \int C_1 dz$$

$$\ln(v) = C_1 z + C_2$$

BC

$$1.) @ z = z_1, x_A = x_{A1}$$

$$\ln(v) = C_1 z + C_2$$

$$2.) @ z = z_2, x_A = x_{A2}$$

$$\ln(1-x_A) = C_1 z + C_2$$

BC1

$$\ln(1-x_{A1}) = C_1 z_1 + C_2$$

BC2

$$\ln(1-x_{A2}) = C_1 z_2 + C_2$$

$$\ln(1-x_{A2}) - \ln(1-x_{A1}) = (C_1 z_2 + C_2) - (C_1 z_1 + C_2)$$

$$\ln \left(\frac{1-x_{A2}}{1-x_{A1}} \right) = C_1 (z_2 - z_1)$$

$$C_1 = \frac{1}{(z_2 - z_1)} \ln \left(\frac{1-x_{A2}}{1-x_{A1}} \right)$$

$$\ln(1-x_A) - \ln(1-x_{A1}) = C_1 z + C_2 - C_1 z_1 - C_2$$

$$\ln \left(\frac{1-x_A}{1-x_{A1}} \right) = C_1 (z - z_1)$$

$$\ln \left(\frac{1-x_A}{1-x_{A1}} \right) = (z - z_1) \frac{1}{(z_2 - z_1)} \ln \left(\frac{1-x_{A2}}{1-x_{A1}} \right)$$

$$\ln \left(\frac{1-x_A}{1-x_{A1}} \right) = \ln \left[\left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{z-z_1}{z_2-z_1}} \right]$$

$$\frac{1-x_A}{1-x_{A1}} = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{z-z_1}{z_2-z_1}}$$

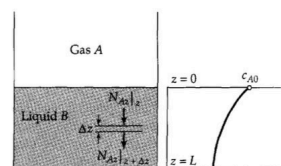


Fig. 18.4-1. Absorption of A by B with a homogeneous reaction in the liquid phase.

$$\rho \left(\frac{\partial \omega_A}{\partial t} + (\mathbf{v} \cdot \nabla) \omega_A \right) = \rho \mathcal{D}_{AB} \nabla^2 \omega_A + r_A \quad (19.1-16)$$

$$\mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} - k_1'' c_A = 0 \quad (18.4-4)$$

Assumptions

Steady State

No Convection

Constant ρ and D_{AB}

Reaction: $A + B \rightarrow AB$ where: $R = k_1'' c_A$

$$\frac{d c_A}{d x} = \frac{d c_A}{d y} = 0$$

$$c_A = \frac{\rho \omega_A}{M_A} = \frac{c_A M_A}{\rho}$$

$$\rho \left(\frac{\partial \omega_A}{\partial t} + (\mathbf{v} \cdot \nabla) \omega_A \right) = \rho \mathcal{D}_{AB} \nabla^2 \omega_A + r_A$$

oess No convection

$$0 = \rho \mathcal{D}_{AB} \nabla^2 \omega_A + r_A$$

$$0 = \rho \mathcal{D}_{AB} \frac{\partial^2 \omega_A}{\partial z^2} + r_A$$

$$0 = \rho \mathcal{D}_{AB} \frac{d}{dz^2} \left(\frac{c_A M_A}{\rho} \right) + r_A$$

$$0 = \rho \frac{M_A}{\rho} \mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + r_A$$

Molar Mass of A Mass of A generated per unit time per unit vol

$$0 = \mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + \frac{r_A}{M_A}$$

$$0 = \mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + R_A$$

$$0 = \mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + k_1'' c_A$$

19B.4. Oxidation of silicon (Fig. 19B.4).¹ A slab of silicon is exposed to gaseous oxygen (species A) at pressure p , producing a layer of silicon dioxide (species B). The layer extends from the surface at $z = 0$, where the oxygen dissolves with concentration $c_{A0} = K p$, to the surface at $z = \delta(t)$, where the oxygen and silicon undergo a first-order reaction with rate coefficient k_1' . The thickness $\delta(t)$ of the growing oxide layer is to be predicted. A quasi-steady-state method is useful here, inasmuch as the advancement of the reaction front is very slow.

(a) First solve the diffusion equation of Eq. 19.1-18, with the term $\partial c_A / \partial t$ neglected, and apply the boundary conditions to obtain

$$c_A = c_{A0} - (c_{A0} - c_{As}) \frac{z}{\delta} \quad (19B.4-1)$$

in which the concentration c_{As} at the reaction plane is as yet unknown.

(b) Next use an unsteady-state molar O_2 balance on the region $0 < z < \delta(t)$ to obtain, with the aid of the Leibniz formula of §C.3,

$$c_{As} \frac{d\delta}{dt} = -\mathcal{D}_{AB} \frac{dc_A}{dz} - k_1' c_{As} \quad (19B.4-2)$$

(c) Now write an unsteady-state molar balance on SiO_2 in the same region to obtain

$$+k_1' c_{As} = \frac{1}{V_B} \frac{d\delta}{dt} \quad (19B.4-3)$$

(d) In Eq. 19B.4-2, evaluate $d\delta/dt$ from Eq. 19B.4-3 and dc_A/dz from Eq. 19B.4-1. This will yield an equation for c_{As} :

$$\frac{k_1' \delta}{\mathcal{D}_{AB}} c_{As} + \left(1 + \frac{k_1' \delta}{\mathcal{D}_{AB}} \right) c_{As} = c_{A0} \quad (19B.4-4)$$

Inserting numerical values into Eq. 19B.4-4 shows that the quadratic term can safely be neglected.¹

(e) Combine Eqs. 19B.4-3 and 19B.4-4 (without the quadratic term) to get a differential equation for $\delta(t)$. Show that this leads to

$$\frac{\delta^2}{2\mathcal{D}_{AB}} + \frac{\delta}{k_1'} = \bar{V}_B c_{A0} t \quad (19B.4-5)$$

which agrees with experimental data.¹ Interpret the result.

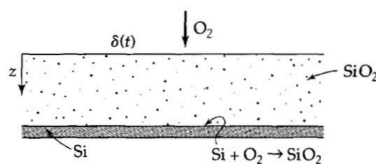


Fig. 19B.4. Oxidation of silicon.

$$\frac{\partial c_A}{\partial t} = \mathcal{D}_{AB} \nabla^2 c_A \quad (19.1-18)$$

$$\frac{\partial c_A}{\partial z} = \mathcal{D}_{AB} \nabla^2 c_A$$

$$N_A(x) = N_A(y) = 0$$

$$N_A = N_A(z)$$

$$0 = \mathcal{D}_{AB} \nabla^2 c_A$$

$$0 = \nabla^2 c_A$$

$$0 = \frac{\partial^2 c_A}{\partial z^2} = \frac{1}{dz^2} c_A = \frac{\partial^2 c_A}{\partial z^2}$$

$$\frac{1}{dz} c_A = \frac{\partial c_A}{\partial z}$$

$$c_A(z) = c_1 z + c_2$$

$$BC: z=0, c_A = c_{A0}$$

$$c_{A0} = c_1(0) + c_2$$

$$c_{A0} = c_2$$

$$BC: z=\delta, c_A = c_{As}$$

$$c_{As} = c_1(\delta) + c_2$$

$$c_{As} = \delta c_1 + c_{A0}$$

$$\frac{c_{As} - c_{A0}}{\delta} = c_1$$

$$c_A(z) = \frac{(c_{As} - c_{A0})}{\delta} z + c_{A0}$$

$$c_A(z) = c_{A0} - \frac{(c_{A0} - c_{As})}{\delta} z$$

b.)

$$\frac{d}{dt} \int_0^{\delta(t)} c_A dz = \int_0^{\delta(t)} \frac{\partial c_A}{\partial t} dz + c_A|_{\delta} \frac{d\delta}{dt}$$

$$\frac{d}{dt} \int_0^{\delta(t)} c_A dz = -(N_{Az}|_{\delta} - N_{Az}|_0) + c_{As} \frac{d\delta}{dt}$$

$$c_{As} \frac{d\delta}{dt} = -k_1' c_{As} - \left(-\mathcal{D}_{AB} \frac{dc_A}{dz} \right)_{z=0}$$

$$c_{As} \frac{d\delta}{dt} = -\mathcal{D}_{AB} \frac{dc_A}{dz} - k_1' c_{As}$$

$$c.) \quad R = \frac{1}{V_B} \left(\frac{dS}{dt} \right)$$

$$K_1'' C_{AS} = \frac{1}{V_B} \frac{dS}{dt}$$

$$d.) \quad C_A(z) = C_{A0} - (C_{A0} - C_{AS}) \frac{z}{\delta}$$

$$- (C_{A0} - C_{AS}) \frac{1}{\delta} = \frac{dC_A}{dz}$$

$$C_{AS} \frac{dS}{dt} = -D_{AB} \frac{dC_A}{dz} - K_1'' C_{AS}$$

$$C_{AS} \left(K_1'' C_{AS} V_B \right) = -D_{AB} \left(\frac{-(C_{A0} - C_{AS})}{\delta} \right) - K_1'' C_{AS}$$

$$V_B K_1'' C_{AS}^2 = -D_{AB} \left(\frac{-(C_{A0} - C_{AS})}{\delta} \right) - K_1'' C_{AS}$$

$$\frac{S}{D_{AB}} V_B K_1'' C_{AS}^2 = C_{A0} - C_{AS} - \frac{S K_1'' C_{AS}}{D_{AB}}$$

$$\boxed{\frac{S V_B K_1''}{D_{AB}} C_{AS}^2 + C_{AS} \left(1 + \frac{S K_1''}{D_{AB}} \right) = C_{A0}}$$

$$e.) \quad \frac{S V_B K_1''}{D_{AB}} C_{AS}^2 + C_{AS} \left(1 + \frac{S K_1''}{D_{AB}} \right) = C_{A0}$$

Ignore

$$C_{AS} = \frac{C_{A0}}{\left(1 + \frac{S K_1''}{D_{AB}} \right)}$$

$$K_1'' C_{AS} = \frac{1}{V_B} \frac{dS}{dt}$$

$$C_{AS} = \frac{1}{K_1'' V_B} \frac{dS}{dt}$$

$$\frac{1}{K_1'' V_B} \frac{dS}{dt} = \frac{C_{A0}}{\left(1 + \frac{S K_1''}{D_{AB}} \right)}$$

$$K_1'' V_B \frac{dt}{dS} = \frac{1}{C_{A0}} + \frac{S K_1''}{D_{AB} C_{A0}}$$

$$\frac{dt}{dS} = \frac{1}{K_1'' V_B C_{A0}} + \frac{S}{V_B D_{AB} C_{A0}}$$

$$\int_0^t dt = \int_0^S \left(\frac{1}{K_1'' V_B C_{A0}} + \frac{S}{V_B D_{AB} C_{A0}} \right) dS$$

$$t = \frac{S}{K_1'' V_B C_{A0}} + \frac{S^2}{2 V_B D_{AB} C_{A0}}$$

$$\boxed{V_B C_{A0} t = \frac{S}{K_1''} + \frac{S^2}{2 D_{AB}}}$$

19D.1. Derivation of the equation of continuity. In §19.1 the species equation of continuity is derived by making a mass balance on a small rectangular volume $\Delta x \Delta y \Delta z$ fixed in space.

(a) Repeat the derivation for an arbitrarily shaped volume element V with a sufficiently smooth fixed boundary S . Show that the species mass balance can be written as

$$\frac{d}{dt} \int_V \rho_A dV = - \int_S (\mathbf{n} \cdot \mathbf{n}_A) dS + \int_V r_A dV \quad (19D.1-1)$$

Use the Gauss divergence theorem to convert the surface integral to a volume integral, and then obtain Eq. 19.1-6.

(b) Repeat the derivation using a region of fluid contained within a surface, each point of which is moving with local mass average velocity.

$$a.) \quad \text{accumulation} = \text{out} - \text{in} + \text{generation}$$

$$\frac{d}{dt} \int_V \rho_A dV = - \int_S (\mathbf{n} \cdot \mathbf{n}_A) dS + \int_V r_A dV$$

$$\int_S (\mathbf{n} \cdot \mathbf{n}_A) dS = \int_V (\nabla \cdot \mathbf{n}_A) dV$$

$$\int_V \frac{\partial \rho_A}{\partial t} dV = - \int_V (\nabla \cdot \mathbf{n}_A) dV + \int_V r_A dV$$

$$b.) \quad n_A = g_A V + J_A$$

$$\frac{d}{dt} \int_{V(t)} \rho_A dV = - \int_{S(t)} (\mathbf{n} \cdot \mathbf{J}_A) dS + \int_{V(t)} r_A dV$$

$$\int_{V(t)} \left(\frac{\partial \rho_A}{\partial t} + \nabla \cdot (\rho_A \mathbf{V}) \right) dV = \int_{V(t)} (-\nabla \cdot \mathbf{J}_A + r_A) dV$$

$$\frac{\partial \rho_A}{\partial t} + \nabla \cdot (\rho_A \mathbf{V}) = -\nabla \cdot \mathbf{J}_A + r_A$$

$$\int_V \left(\frac{\partial \rho_A}{\partial t} + \nabla \cdot \mathbf{n}_A - \Gamma_A \right) dV = 0$$

$$\frac{\partial \rho_A}{\partial t} + (\nabla \cdot \mathbf{n}_A) = \Gamma_A$$

$$\frac{D \rho_A}{Dt} = \frac{\partial \rho_A}{\partial t} + \mathbf{v} \cdot \nabla \rho_A$$

$$\nabla \cdot (\rho_A \mathbf{v}) = \rho_A (\nabla \cdot \mathbf{v}) + \mathbf{v} \cdot \nabla \rho_A$$

$$\frac{D \rho_A}{Dt} + \rho_A (\nabla \cdot \mathbf{v}) = -\nabla \cdot \mathbf{J}_A + \Gamma_A$$